FULL NAME: ______________________ **BLOCK:** __________

Radical Equations With Rational Exponents + DOK-3 Modeling

Lesson 5-4 · Period 2 · Teacher Edition

OBJECTIVES

Math Objective	Solve equations with rational exponents by converting to radical form and raising both sides to the reciprocal power; apply rational-exponent models to real-world half-life and decay problems.
Language Objective	Translate a real-world half-life scenario into an equation with a rational exponent, compute, and interpret the result in context.
Essential Understanding	Rational exponents are the bridge between algebraic power operations and real-world exponential/radical modeling — the half-life formula $y = a \cdot (1/2)^{(t)/(h)}$ uses fractional powers directly.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Solve rational-exponent equations; model real-world decay with fractional-power formulas.
Essential Question	How do rational-exponent equations model real-world decay, and when do extraneous solutions arise?
Materials	Student packet, pencil, calculator for Q#41. Teacher models Ex 4; DOK-3 Practice #41 is the topic's capstone modeling task.

[PIN] PRE-FLIGHT — P2 IS THE DOK-3 SPINE DAY

Today's conceptual core is Practice #41 (Half-Life Model). Students must (a) identify $a = 50$, $h = 5$, $t = 16$, (b) compute exponent $16/5 = 3.2$, (c) evaluate $50 \cdot (0.5)^{3.2} \approx 5.44$ mL, (d) interpret: 'about 5.44 mL of the drink remains after 16 hours.' This is the single most important item in Topic 5 — all three assessment items chain off this skill.
Pace: Try-It 4 (4 min) → #12 (4 min) → #33 (4 min) → #34 (4 min) → #41 (12 min). ∼35 min total Explore. (Adjust: Try-It 4 + #12 = 8 min block, #33 + #34 = 8 min block.)
If running tight: cut #33 or #34. Never cut #41.

Tag: bridge from P1

Do Now **DOK:** 1 **Minutes:** 5

Teacher does

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student for radical isolation and solution check.

Students do

- Solve by isolating the radical and raising to the reciprocal power.
- Check for extraneous solutions.

Questions to ask

- What operation undoes a cube root?
- Did you verify by substituting back into the original?

Adult role
Solo teacher: circulate silently. No hints.

Do Now (5-4-savvas-q21)
Solve each radical equation. ($\sqrt[3]{x} + 8 = 13$).

[TARGET] SENTENCE FRAME

"I isolate the radical, giving ___. I raise both sides to the power ___, getting ___. I check by substituting back: the left side equals ___ and the right side equals ___. The solution is ___."

[CHECK] ANSWER KEY

Answer key (from source): (x=125)

Tag: teacher models Example 4

Launch **DOK:** 2 **Minutes:** 12

Teacher does

- Project Example 4. Think aloud: rewrite the rational exponent as a radical, then raise both sides to the reciprocal power.
- "The exponent m/n means I raise to the n/m to undo it — the reciprocal flips the fraction."
- Flag the Rational-Exponent Rules card — print it in student minds.
- 30-sec pause: "What is the reciprocal of the exponent? Can an extraneous solution appear?" (partner check)
- Do NOT solve Try-Its or Practice items — release at minute 17.

Students do

- Watch Example 4 silently.
- During pause: state the reciprocal exponent to partner.
- Copy Rules card into margin.
- Note the half-life formula — this comes back in #41.

Questions to ask

- What must I do BEFORE raising both sides to the reciprocal power?
- Why do I need to check my solution even after correct algebra?
- In $y = a \cdot (0.5)^{(t)/(h)}$, which value is the half-life? (h)

Adult role
Project. Think aloud. Release promptly at minute 17.

[BOOK] RATIONAL-EXPONENT RULES (keep printed on your page)

RATIONAL-EXPONENT RULES
1. Rewrite $x^{m/n}$ as $(\sqrt[n]{x})^m$ — bridge to radicals you already know.
2. To solve $(\text{expr})^{m/n} = k$, raise both sides to the reciprocal power n/m.
3. Simplify, then CHECK — even rational-exponent equations can have extraneous solutions when the denominator of the exponent is EVEN.
4. For real-world decay $y = a \cdot (0.5)^{(t)/(h)}$: half-life h tells you the exponent's denominator; plug in t, evaluate.

[WARN] EXTRANEOUS SOLUTIONS ARISE WHEN THE DENOMINATOR IS EVEN

$(\text{expr})^{m/n} = k \rightarrow$ raise both sides to n/m
If n is EVEN, squaring can introduce solutions that don't satisfy the original equation.
ALWAYS substitute back into the ORIGINAL equation to verify.

Example 4 (5-4-savvas-example-4-lesson-5-4)

Solve Equations With Rational Exponents
A. What are the solutions to the equation $(x^2+5x+5)^{5/2} = 1$**?**

$$\begin{aligned}
(x^2+5x+5)^{5/2} &= 1\\
\left((x^2+5x+5)^{5/2}\right)^{2/5} &= (1)^{2/5}\\
x^2+5x+5 &= 1\\
x^2+5x+4 &= 0\\
(x+4)(x+1) &= 0\\
x+4 = 0 \quad &\text{or} \quad x+1=0\\
x=-4 \quad &\text{or} \quad x=-1
\end{aligned}$$

Check for extraneous solutions.

$$\begin{aligned}
((-4)^2+5(-4)+5)^{5/2} &\stackrel{?}{=} 1\\
(16-20+5)^{5/2} &\stackrel{?}{=} 1\\
(1)^{5/2} &\stackrel{?}{=} 1\\
1 &= 1 \quad \checkmark
\end{aligned}
\qquad
\begin{aligned}
((-1)^2+5(-1)+5)^{5/2} &\stackrel{?}{=} 1\\
(1-5+5)^{5/2} &\stackrel{?}{=} 1\\
(1)^{5/2} &\stackrel{?}{=} 1\\
1 &= 1 \quad \checkmark
\end{aligned}$$

This equation has two solutions, $x=-4$ and $x=-1$. There are no extraneous solutions.
B. What is the solution to $(x+18)^{3/2} = (x-2)^3$**?**

$$\begin{aligned}
(x+18)^{3/2} &= (x-2)^3\\
\left((x+18)^{3/2}\right)^{2/3} &= \left((x-2)^3\right)^{2/3}\\
x+18 &= (x-2)^2\\
x+18 &= x^2-4x+4\\
x^2-5x-14 &= 0\\
(x+2)(x-7) &= 0\\
x+2=0 \quad &\text{or} \quad x-7=0\\
x=-2 \quad &\text{or} \quad x=7
\end{aligned}$$

Check for extraneous solutions.

$$(-2+18)^{3/2} \stackrel{?}{=} (-2-2)^3 \qquad (7+18)^{3/2} \stackrel{?}{=} (7-2)^3$$
$$16^{3/2} \stackrel{?}{=} (-4)^3 \qquad 25^{3/2} \stackrel{?}{=} 5^3$$
$$64 \neq -64 \quad \text{X} \qquad 125 = 125 \quad \checkmark$$

So 7 is the only solution to the equation, and -2 is an extraneous solution.

[CHECK] WORKED CONCLUSION

A. This equation has two solutions, $x = -4$ and $x = -1$. There are no extraneous solutions.
B. So 7 is the only solution to the equation, and -2 is an extraneous solution.

[MIC] TEACHER SCRIPT

1. "Watch me solve an equation with a rational exponent. I rewrite $x^{m/n}$ as the nth root of x, raised to the m."
2. "Step 1: Isolate the expression with the rational exponent on one side."
3. "Step 2: Raise BOTH sides to the reciprocal power n/m to eliminate the exponent."
4. "Step 3: Simplify and solve for the variable."
5. "Step 4: CHECK by substituting back into the original — even perfect algebra can produce extraneous solutions when the denominator is even."
6. "Today's DOK-3 task applies this exact skill to a real-world half-life model. The exponent t/h IS a rational exponent."
7. Release to Explore.

Tag: TPS + DOK-3 driver

Explore **DOK:** 2–3 **Minutes:** 33

Teacher does

- 4 laps across 33 min.
- Lap 1 (8 min): Try-It 4 + #12 — rational-exponent equations. Press pairs to write the reciprocal power explicitly.
- Lap 2 (8 min): #33 + #34 — rational-exponent equations. Watch for pairs who skip the check step.
- Lap 3 (12 min): [STAR] #41 Half-Life Model. Students identify a, h, t; compute exponent; evaluate; interpret.
- Do NOT give #41 answer. Pairs present computation and interpretation at Share.

Students do

- 5 items in order: Try-It 4 → #12 → #33 → #34 → #41.
- For #41: BEFORE computing, identify $a = 50$ (start amount), $h = 5$ (half-life period in hours), $t = 16$ (elapsed hours).
- Justify with sentence frame.

Questions to ask

- What is the reciprocal of the exponent? Write it before you raise both sides.
- #12: does the denominator of the exponent tell you anything about extraneous solutions?
- #33 / #34: did you check your solution in the original equation?
- #41: what does $a = 50$ represent? What does $h = 5$ represent?
- #41: compute 16/5. What decimal is that? Now evaluate $(0.5)^{3.2}$.
- #41: interpret your answer — what does 5.44 mL mean in context?

Adult role
Solo teacher: 4 laps. Press INTERPRETATION on #41 — decimal answer alone is not enough.

Work with a partner. Move in order. Practice #41 (Half-Life Model) is the big one — plan ∼12 min for it.

Try It 4 (5-4-savvas-try-it-4-lesson-5-4)
Solve each equation.
a. $(x^2 - 3x - 6)^{3/2} - 14 = -6$
b. $(x - 8)^{3/2} = (10 - x)^3$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): x = 10, -7

Practice #12 (5-4-savvas-q12)
$(3x + 2)^{2/3} = 4$

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

[CHECK] DERIVED TEACHER ANSWER

No source answer was present in the registry/docx build. Solving $(3x + 2)^{2/3} = 4$ gives $x = 2$ or $x = -\frac{10}{3}$.

Practice #33 (5-4-savvas-q33)
Solve. $\left(0.5(x^2 + 5x + 136)^{(2)/(3)} = 50\right)$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): (x=27) and (x=-32)

Practice #34 (5-4-savvas-q34)
Solve. $\left(2(x^2 - 12x - 4)^{(1)/(2)} - 3 = 15\right)$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): (x=-5) and (x=17)

[STAR] DOK-3 HALF-LIFE MODEL — Practice #41 (Topic 5 spine) (5-4-savvas-q41)

Make Sense and Persevere The half-life of a certain type of soft drink is 5 h. If you drink 50 mL of this drink, the formula $y = 50(0.5)^{(t)/(5)}$ tells the amount of the drink left in your system after t hours. How much of the soft drink will be left in your system after 16 hours?

[STAR] DOK-3 HALF-LIFE MODEL — Practice #41 (Topic 5 spine)

This is the single most important item in the topic. Students should (a) identify $a = 50$, $h = 5$, $t = 16$, (b) compute exponent $16/5 = 3.2$, (c) evaluate $50 \cdot (0.5)^{3.2} \approx 5.44$ mL, (d) interpret: 'about 5.44 mL of the drink remains after 16 hours.' All three assessment items chain off this skill.

[CHECK] ANSWER KEY

Answer key (from source): about 5.44 mL

Tag: academic conversation + P3 bridge

Share / Summary **DOK:** 2 **Minutes:** 5

Teacher does

- Cold-call 1 pair to present their Half-Life computation and interpretation using sentence frame.
- Class signals thumbs. Write the half-life formula $y = a \cdot (0.5)^{(t)/(h)}$ on board with a, h, t labeled.
- P3 bridge: "Next period we apply these skills to assessment-critical items — bring your check-for-extraneous habit."

Students do

- Presenting pair reads their computation and interpretation.
- Rest of class signals and writes takeaway in margin.

Questions to ask

- Summarize: what does the exponent t/h represent in the half-life formula?
- Does a rational exponent always produce a unique solution? (No — check for extraneous.)

Adult role

Cold-call a pair that struggled on #41 — hold them accountable to the full interpretation step.

Sentence frames:

Pair share (Half-Life Model): "This equation has exponent ___/___, so I raise both sides to the power ___/___. After simplifying I get ___. In the half-life formula, $a =$ ___ (start), $h =$ ___ (half-life period), $t =$ ___ (elapsed time), giving $y \approx$ ___."

Whole class: one pair shares their computation and interpretation. Class signals thumbs up/sideways.

[BRIDGE] BRIDGE TO P3

Next period we apply rational-exponent skills to assessment-critical items. Bring your check-for-extraneous habit.

Tag: summary recap (not graded)

Exit Ticket **DOK:** 1–2 **Minutes:** 3

Teacher does

- Collect at door.
- Scan stem 3 ($50 \rightarrow 5.44$ mL). Plan P3 Do Now if many students got the computation wrong or skipped the interpretation.

Students do

- 4 sentence stems, honest.

Questions to ask

- Did students correctly compute $(0.5)^{3.2}$ and interpret the result in context?

Adult role
Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

To solve $(\text{expr})^{m/n} = k$ I raise both sides to power ___.
Half-life tells me the ___ of the exponent.
After 16 hours, 50 mL of soft drink decays to ___ mL.
One biggest thing I learned today: ______________________________

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #19 is a conceptual extraneous-solution argument that extends the half-life skill.

Optional #1 (Savvas Practice, extraneous-solution argument)
Higher Order Thinking Some, but not all, equations with rational exponents have extraneous solutions. What is the relationship between the exponents and the possibility of having extraneous solutions for equations with rational exponents? Explain your reasoning.

[CHECK] ANSWER KEY

Answer key (from source): Only if the denominator of the rational exponent is even can there be extraneous solutions. Taking an even root of a negative number gives an extraneous solution, so if the root isn't even, there can't be an extraneous solution.

[CLOCK] PERIOD A / PERIOD F — 65-MIN CLASS NOTES

Both sections should have 65 min for P2. Use the extra 10 min on Explore #41 (Half-Life Model) — press students to write the full interpretation sentence, not just the decimal.
If finished early: hand out Practice #19 (extraneous-solution argument) as fast-finisher.

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Rational-Exponent Rules card as a sticker/cut-out.
- For #41, provide the formula $y = a \cdot (0.5)^{(t)/(h)}$ pre-labeled with $a = 50$, $h = 5$, $t = 16$.
- Accept verbal interpretation in lieu of written sentence.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: "reciprocal power" = flip the exponent fraction; "half-life" = time for half the amount to remain; "extraneous solution" = an answer the algebra gives but the original equation rejects.
- "Half-Life Model" = a formula showing how something decreases by half every h hours.
- Pair ELL students with English-dominant partners for TPS.